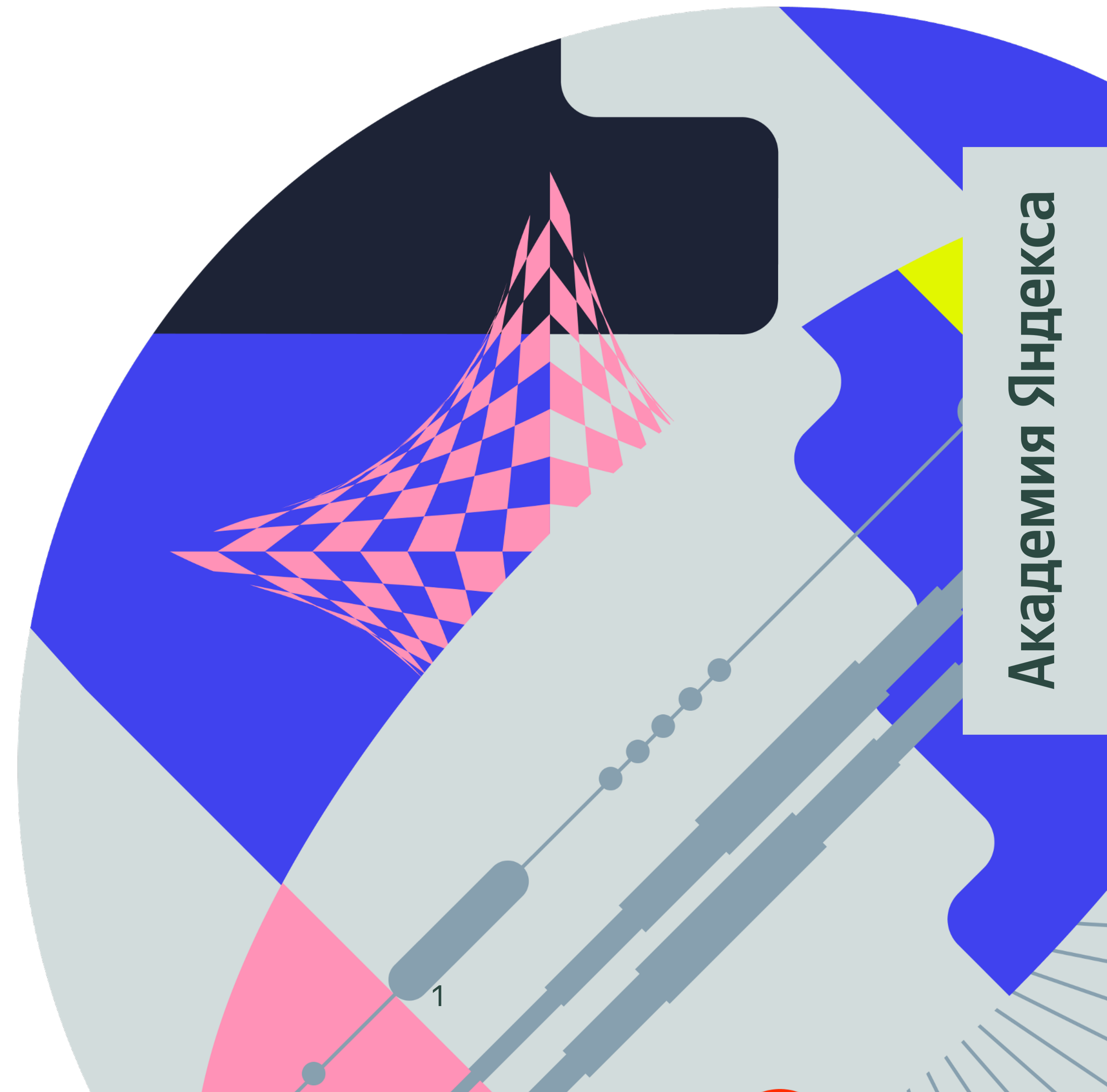


Visual Generative Modeling 2026

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Yandex Research



Lecture 3 | Numerical solvers

1. **Recap from the previous lectures**
2. Basics of numerical methods for ODEs
3. **DPM-solver (DDIM, Euler, Heun)**
4. SDE solvers

1. Recap | Probabilistic DDPM (VAE perspective)

Forward process

1. Sequence of random variables

$$\{x_j\}_{j=1}^T, x_0 \sim q(x_0), x_T \sim \text{noise}$$

2. Proposal distribution

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t | \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$$

3. Joint distribution

$$q(x_0, \dots, x_T) = q(x_0) \prod_{t=1}^T q(x_t | x_{t-1})$$

1. Recap | Probabilistic DDPM (VAE perspective)

Forward process

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 $\{x_j\}_{j=1}^T, x_0 \sim q(x_0), x_T \sim \text{noise}$

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3. Joint distribution
 $q(x_0, \dots, x_T) = q(x_0) \prod_{t=1}^T q(x_t | x_{t-1})$

Reverse process

1. Sequence of random variables
 $\{x_j\}_{j=T}^1, x_0 \sim p_\theta(x_0), x_T \sim \text{noise}$

2. Proposal distribution
 $p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1} | \mu_\theta(x_t, t), \sigma_t^2 I)$

3. Joint distribution
 $p_\theta(x_T, \dots, x_0) = p_\theta(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t)$

1. Recap | Probabilistic DDPM (VAE perspective)

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$$\{x_j\}_{j=1}^T, x_0 \sim q(x_0), x_T \sim \text{noise}$$

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Reverse process

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3. Joint distribution

$$p_\theta(x_T, \dots, x_0) = p_\theta(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t)$$

Training

???

1. Recap | Probabilistic DDPM (VAE perspective)

Forward process

1. Sequence of random variables
 $\{x_j\}_{j=1}^T, x_0 \sim q(x_0), x_T \sim \text{noise}$

2. Proposal distribution
 $q(x_t | x_{t-1}) = \mathcal{N}(x_t | \sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$

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 $q(x_0, \dots, x_T) = q(x_0) \prod_{t=1}^T q(x_t | x_{t-1})$

Reverse process

1. Sequence of random variables
 $\{x_j\}_{j=T}^1, x_0 \sim p_\theta(x_0), x_T \sim \text{noise}$

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 $p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1} | \mu_\theta(x_t, t), \sigma_t^2 I)$

3. Joint distribution
 $p_\theta(x_T, \dots, x_0) = p_\theta(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t)$

Training

$$\theta^* = \operatorname{argmax}_\theta \mathbb{E}_{q(x_0)} \log p_\theta(x_0)$$

1. Recap | SDE/ODE perspective

Forward process

$$\text{SDE : } d\boldsymbol{x} = \boldsymbol{f}(t)\boldsymbol{x}dt + \boldsymbol{g}(t)d\boldsymbol{w}_t$$

Solution???

1. Recap | SDE/ODE perspective

Forward process

$$\text{SDE : } d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}_t$$

Solution???

$$p_t(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x} \mid m(t), v(t)\right)$$

^{*}
$$\begin{aligned} dm_t &= f(t)m_t dt, \quad m_0 = \mathbf{x}_0 \\ dv_t &= (2f(t)v_t + g^2(t)) dt, \quad v_0 = 0 \end{aligned}$$

1. Recap | SDE/ODE perspective

Forward process

$$\text{SDE : } d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}_t$$

Solution???

$$p_t(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x} \mid m(t), v(t)\right)$$

$$dm_t = f(t)m_t dt, m_0 = \mathbf{x}_0$$

$$dv_t = (2f(t)v_t + g^2(t)) dt, v_0 = 0$$

Reverse process

$$\text{SDE : } d\mathbf{x} = \left[f(t)\mathbf{x} - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt + g(t)d\mathbf{w}_t,$$

$$\text{ODE : } d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2}g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$

Solution???

1. Recap | SDE/ODE perspective

Forward process

$$\text{SDE : } d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}_t$$

Solution???

$$p_t(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x} \mid m(t), v(t)\right)$$

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Reverse process

$$\text{SDE : } d\mathbf{x} = \left[f(t)\mathbf{x} - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt + g(t)d\mathbf{w}_t,$$

$$\text{ODE : } d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2}g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$

Solution???

Only numerically

Training

$$\mathbb{E}_{\mathbf{x}_0, t, \mathbf{x}} \left\| s_{\theta}(\mathbf{x}, t) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}) \right\|^2$$

1. Recap | SDE/ODE perspective

Main entities

$$d\boldsymbol{x} = \left[f(t)\boldsymbol{x} - \frac{1}{2}g(t)^2 \nabla_{\boldsymbol{x}} \log p_t(\boldsymbol{x}) \right] dt$$

$$p_t(\boldsymbol{x} \mid \boldsymbol{x}_0) = \mathcal{N} \left(\boldsymbol{x} \mid m(t), v(t) \right)$$

1. Recap | SDE/ODE perspective

Main entities

$$d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2}g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$

$$p_t(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N} \left(\mathbf{x} \mid m(t), v(t) \right)$$

Path 1. $m(t), v(t) \rightarrow f(t), g(t)$

$$\mathbf{x} = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon} \rightarrow d\mathbf{x} = \frac{1}{2\alpha_t} \left[\mathbf{x} + \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] d\alpha_t$$

1. Recap | SDE/ODE perspective

Main entities

$$d\mathbf{x} = \left[f(t)\mathbf{x} - \frac{1}{2}g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$$

$$p_t(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N} \left(\mathbf{x} \mid m(t), v(t) \right)$$

Path 1. $m(t), v(t) \rightarrow f(t), g(t)$

$$\mathbf{x} = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon} \rightarrow d\mathbf{x} = \frac{1}{2\alpha_t} \left[\mathbf{x} + \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] d\alpha_t$$

Path 2. $f(t), g(t) \rightarrow m(t), v(t)$

$$f(t) = \frac{1}{2\alpha_t} \frac{d\alpha_t}{dt}, g^2(t) = -\frac{1}{\alpha_t} \frac{d\alpha_t}{dt} \rightarrow \mathbf{x} = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}$$

Noising process $\longrightarrow$ ODE (SDE) $\longrightarrow$ Training of NN $\longrightarrow$ Inference using solver

	Noising process	ODE
General case	$\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \mathbf{z},$ $t \in [0, T], \mathbf{z} \sim \mathcal{N}(0, 1)$	$d\mathbf{x} = \left[\mathbf{x} f(t) - \frac{1}{2} g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt$ $f(t) = \frac{d \log \alpha_t}{dt}, g^2(t) = \frac{d \sigma_t^2}{dt} - 2 \frac{d \log \alpha_t}{dt} \sigma_t^2$
VP	$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \mathbf{z}$ $t \in [0, T], \mathbf{z} \sim \mathcal{N}(0, 1)$	$d\mathbf{x} = \frac{1}{2\alpha_t} [\mathbf{x} + \nabla_{\mathbf{x}} \log p_t(\mathbf{x})] d\alpha_t$
VE	$\mathbf{x}_t = \mathbf{x}_0 + t\mathbf{z}$ $t \in [0, T], \mathbf{z} \sim \mathcal{N}(0, 1)$	$d\mathbf{x} = -t \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) dt$
FM (RF)	$\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{z}$ $t \in [0, 1], \mathbf{z} \sim \mathcal{N}(0, 1)$	$d\mathbf{x} = -\frac{1}{1 - t} [\mathbf{x} + t \nabla_{\mathbf{x}} \log p_t(\mathbf{x})] dt$ $d\mathbf{x} = v(\mathbf{x}, t) dt$

1. $\mathbb{E}_{\mathbf{x}_0, t, \mathbf{x}_t} \ \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) - \boldsymbol{\epsilon}\ $	2. $\mathbb{E}_{\mathbf{x}_0, t, \mathbf{x}_t} \ s_{\theta}(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)\ $	3. $\mathbb{E}_{\mathbf{x}_0, t, \mathbf{x}_t} \ \mathbf{x}_{\theta}(\mathbf{x}_t, t) - \mathbf{x}_0\ $
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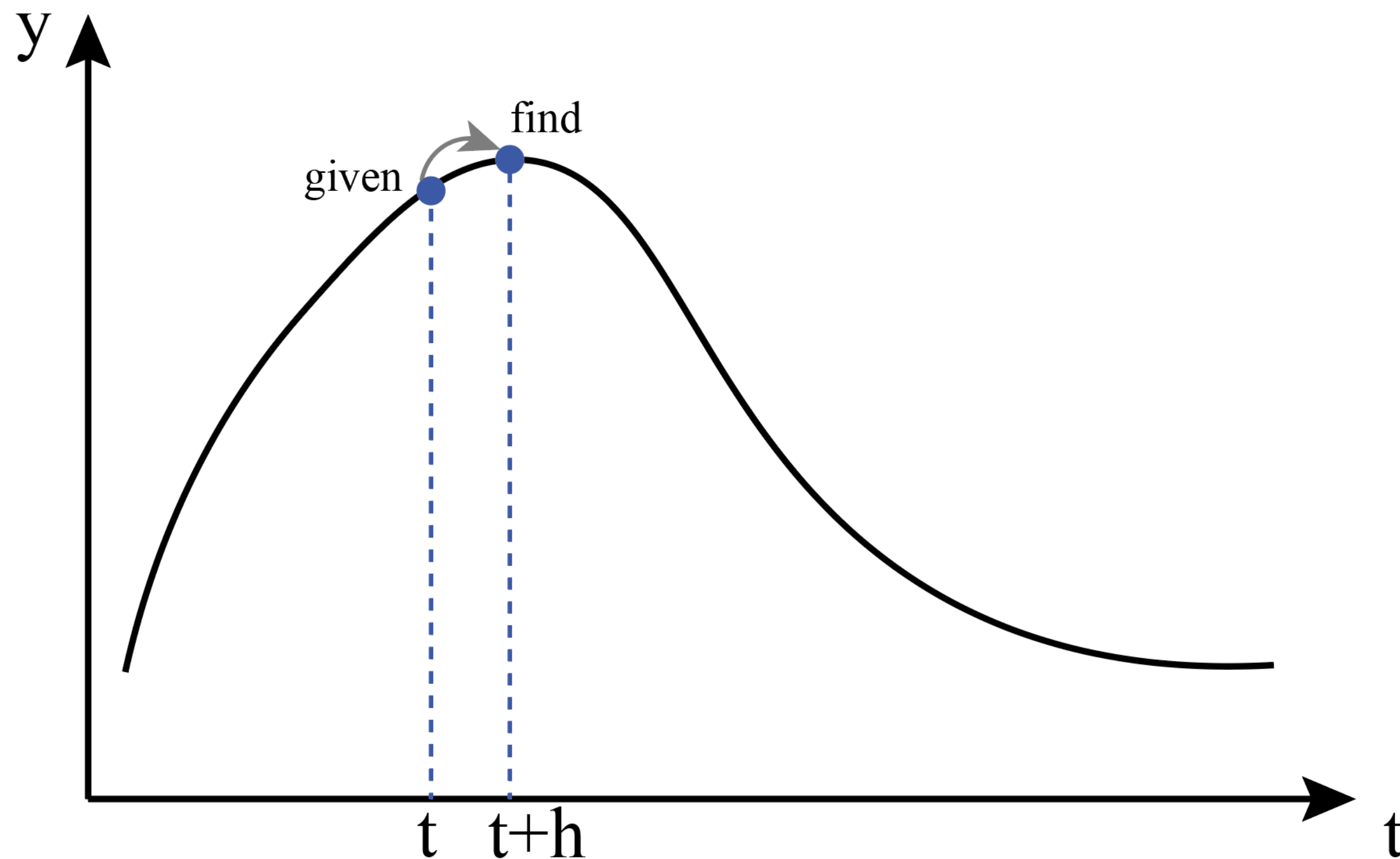
2. Basics of numerical solution of ODEs. Intuition

Cauchy problem

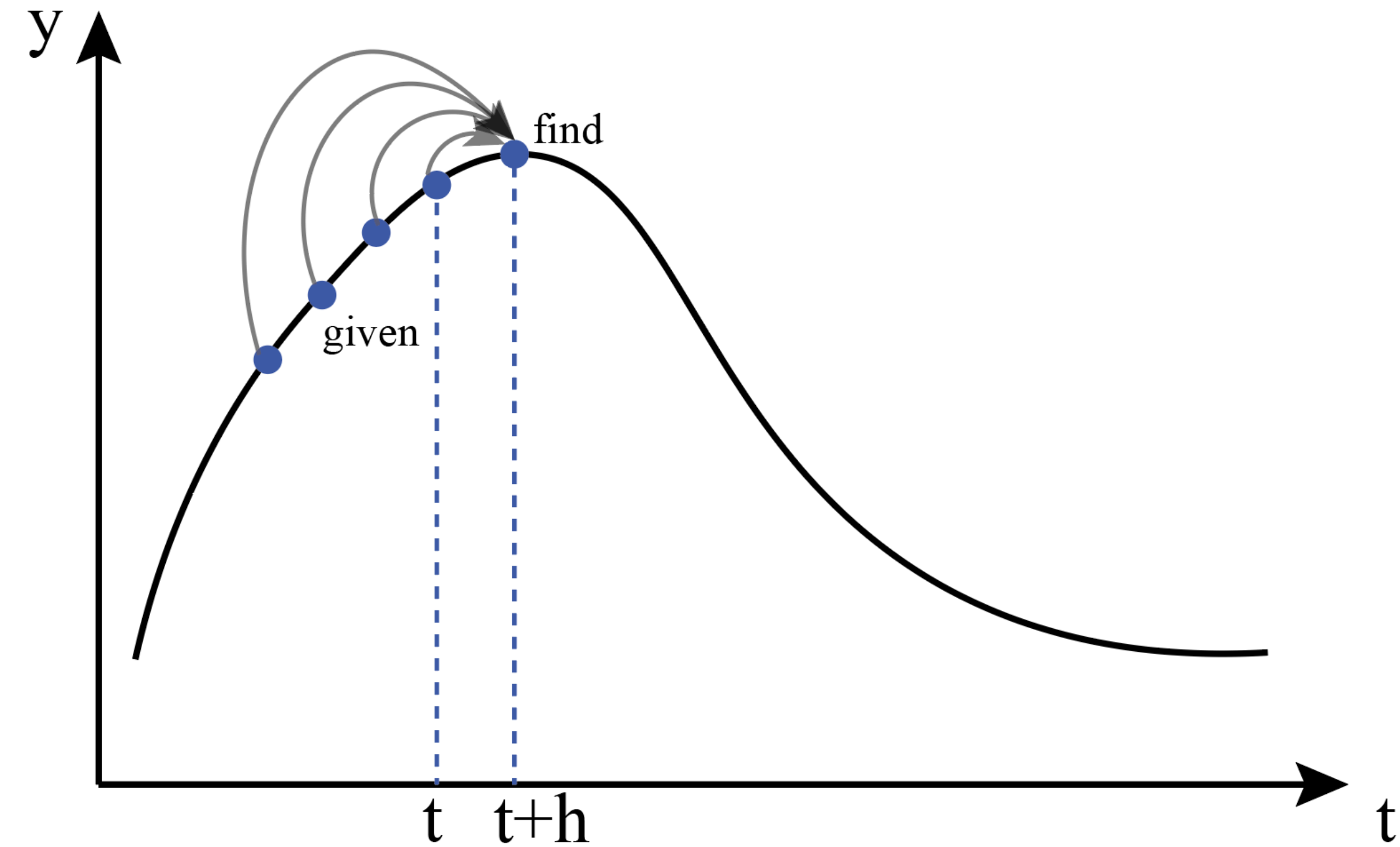
$$dy = f(t, y(t))dt, y(0) = y_0$$

Numerical solvers $[t_0, \dots, t_N], h = t_{i+1} - t_i$

Singlestep $y_{i+1} = y_i + \Phi(y_i, h)$



Multistep $y_{i+1} = y_i + \Phi(y_i, y_{i-1}, \dots, h)$



Singlestep $y_{i+1} = y_i + \Phi(y_i, h)$

Multistep $y_{i+1} = y_i + \Phi(y_i, y_{i-1}, \dots, h)$

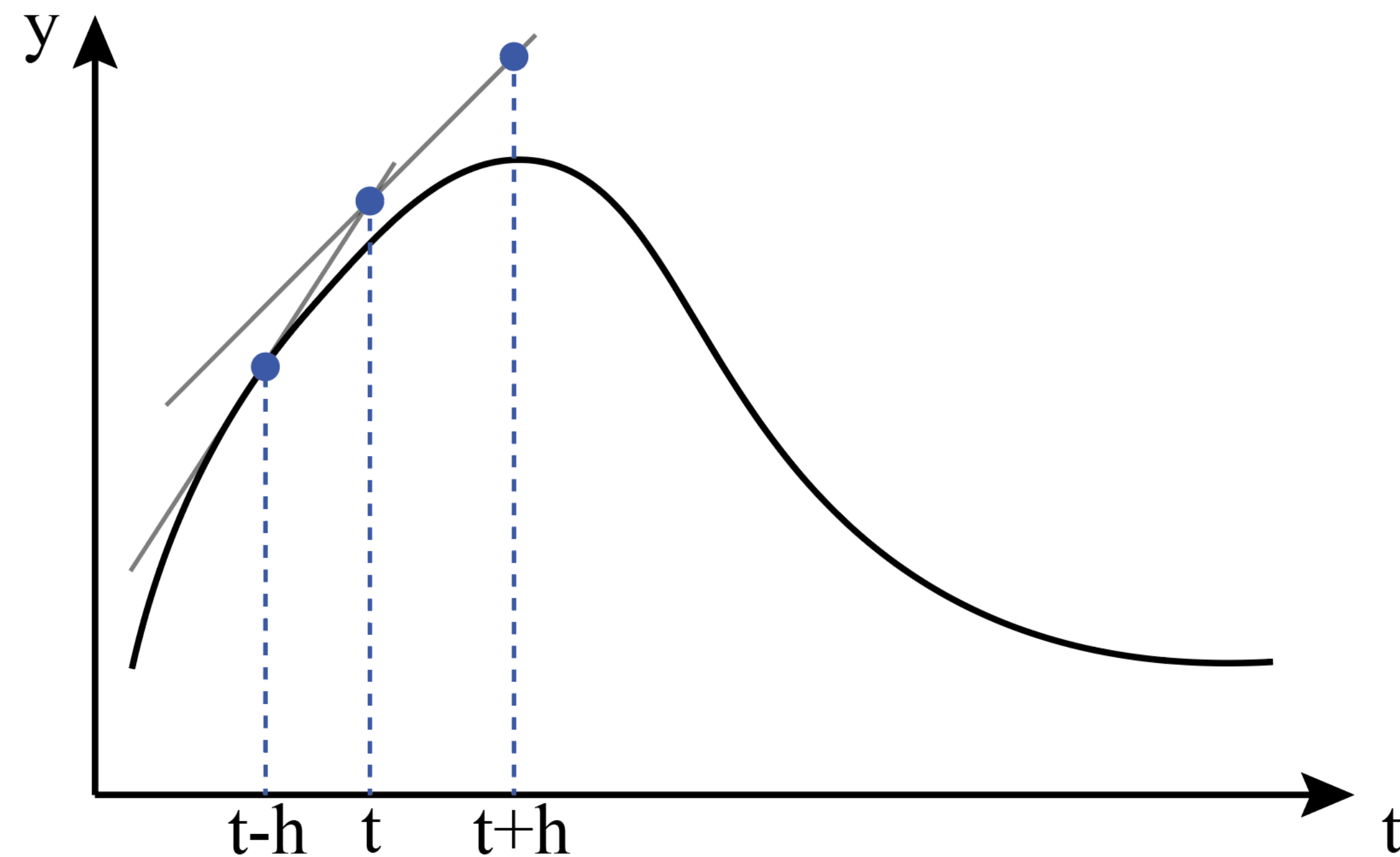
Implicit $y_{i+1} = y_i + \Phi(y_{i+1}, y_i, h)$

Implicit $y_{i+1} = y_i + \Phi(y_{i+1}, y_i, y_{i-1}, \dots, h)$

Examples:

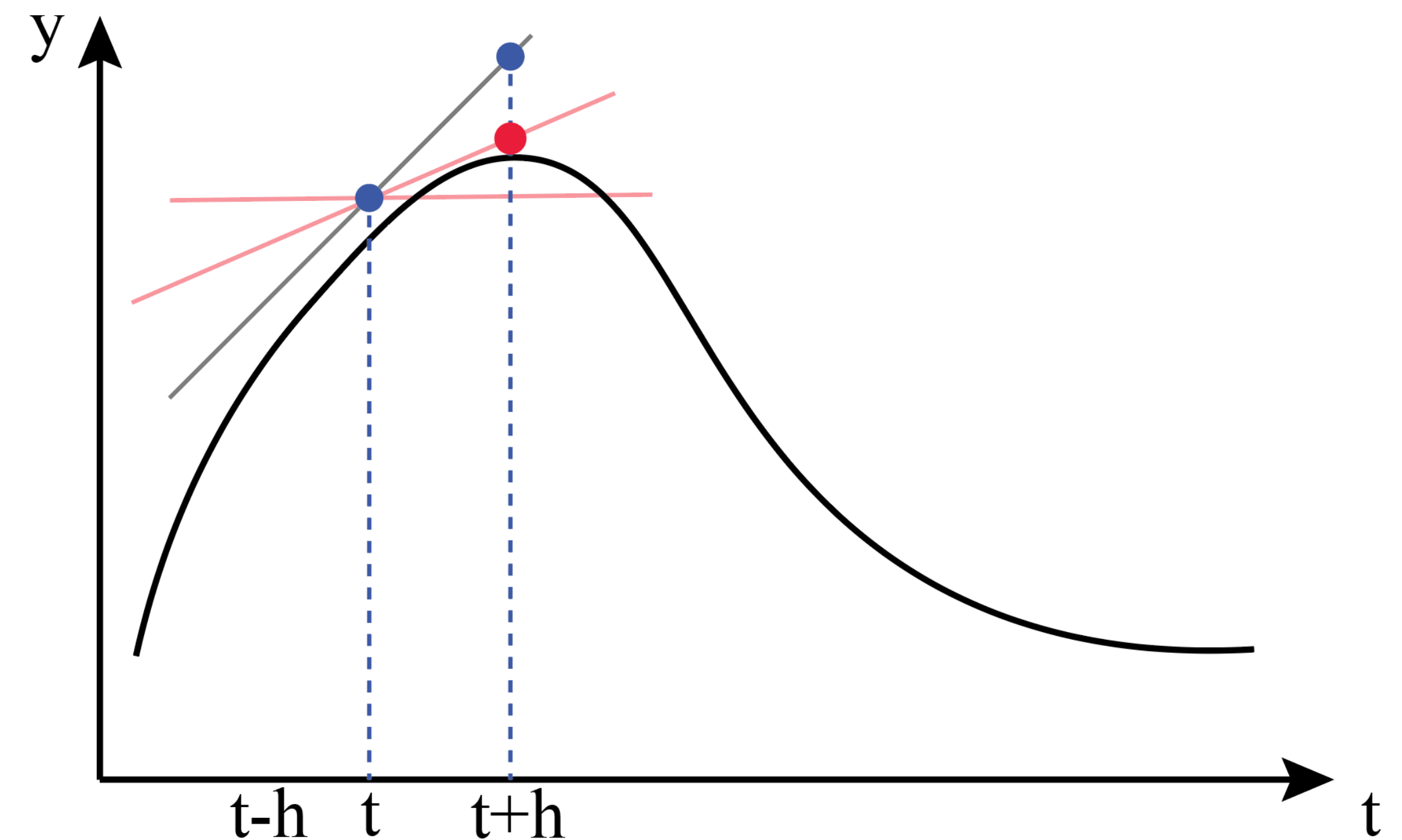
* Euler (singlestep, explicit)

$$y_{i+1} = y_i + hf(y_i, t_i)$$



* Trapezoidal rule (singlestep, implicit)

$$y_{i+1} = y_i + \frac{h}{2} (f(y_i, t_i) + f(y_{i+1}, t_{i+1}))$$



How to understand how accurate is your solver?

1. Local truncation error (we assume the previous steps are exact)

$$\tau_{i+1} = \|y_{i+1} - y(t_{i+1})\|$$

2. Global truncation error (accumulative error after N steps)

$$e_N = \|y_N - y(t_N)\| \quad e_N \sim N\tau \sim \frac{\tau}{h}$$

Solver converges if:

1. LTE $\rightarrow 0$, as $h \rightarrow 0$

2. It is stable

small perturbations do not grow exponentially.

2. Basics of numerical solution of ODEs. Explicit multistep/singlestep methods

Taylor Series Methods (multistep/singlestep)

$$y_{i+1} = y_i + \int_{t_i}^{t_{i+1}} f(t, y(t)) dt$$

$$f(t, y(t)) = \sum_{k=0} \frac{h^k}{k!} f^{(k)}(t_i, y_i)$$

2. Basics of numerical solution of ODEs. Explicit multistep/singlestep methods

Taylor Series Methods (multistep/singlestep)

$$y_{i+1} = y_i + \int_{t_i}^{t_{i+1}} f(t, y(t)) dt$$

$$f(t, y(t)) = \sum_{k=0} \frac{h^k}{k!} f^{(k)}(t_i, y_i)$$

Euler = Taylor Series Methods, singlestep, 1st order

$$y_{i+1} = y_i + \int_{t_i}^{t_{i+1}} f(t, y(t)) dt = y_i + (t_{i+1} - t_i) f(t_i, y_i) + O(h^2)$$

$$f(t, y(t)) = f(t_i, y_i) + O(h)$$

Heun = Taylor Series Methods, singlestep, 2nd order

$$y_{i+1} = y_i + \int_{t_i}^{t_{i+1}} f(t, y(t)) dt = y_i + (t_{i+1} - t_i)f(t_i, y_i) + \frac{1}{2}(t_{i+1} - t_i)^2 f'(t_i, y_i) + O(h^3)$$

$$f(t, y(t)) = f(t_i, y_i) + hf'(t_i, y_i) + O(h^2)$$

Heun = Taylor Series Methods, singlestep, 2nd order

$$y_{i+1} = y_i + \int_{t_i}^{t_{i+1}} f(t, y(t)) dt = y_i + (t_{i+1} - t_i)f(t_i, y_i) + \frac{1}{2}(t_{i+1} - t_i)^2 f'(t_i, y_i) + O(h^3)$$

$$f(t, y(t)) = f(t_i, y_i) + hf'(t_i, y_i) + O(h^2)$$

How to approximate $f'(t_i, y_i)$?

$$f'(t_i, y_i) \approx \frac{f(t_{i+\delta_i}, y_{i+\delta_i}) - f(t_i, y_i)}{t_{i+\delta_i} - t_i}$$

Heun : $\delta = 1$, $y_{i+1}^* = y_i + (t_{i+1} - t_i)f(t_i, y_i)$

Heun = Taylor Series Methods, singlestep, 2nd order

$$\text{Heun : } \delta = 1, y_{i+1}^* = y_i + (t_{i+1} - t_i)f(t_i, y_i)$$

$$y_{i+1} = y_i + (t_{i+1} - t_i)f(t_i, y_i) + \frac{1}{2}(t_{i+1} - t_i)^2 \frac{f(t_{i+1}, y_{i+1}^*) - f(t_i, y_i)}{t_{i+1} - t_i} + O(h^3)$$

$$y_{i+1} = y_i + \frac{(t_{i+1} - t_i)}{2} \left(f(t_i, y_i) + f(t_{i+1}, y_{i+1}^*) \right) + O(h^3)$$

Taylor Series Methods, multistep ???

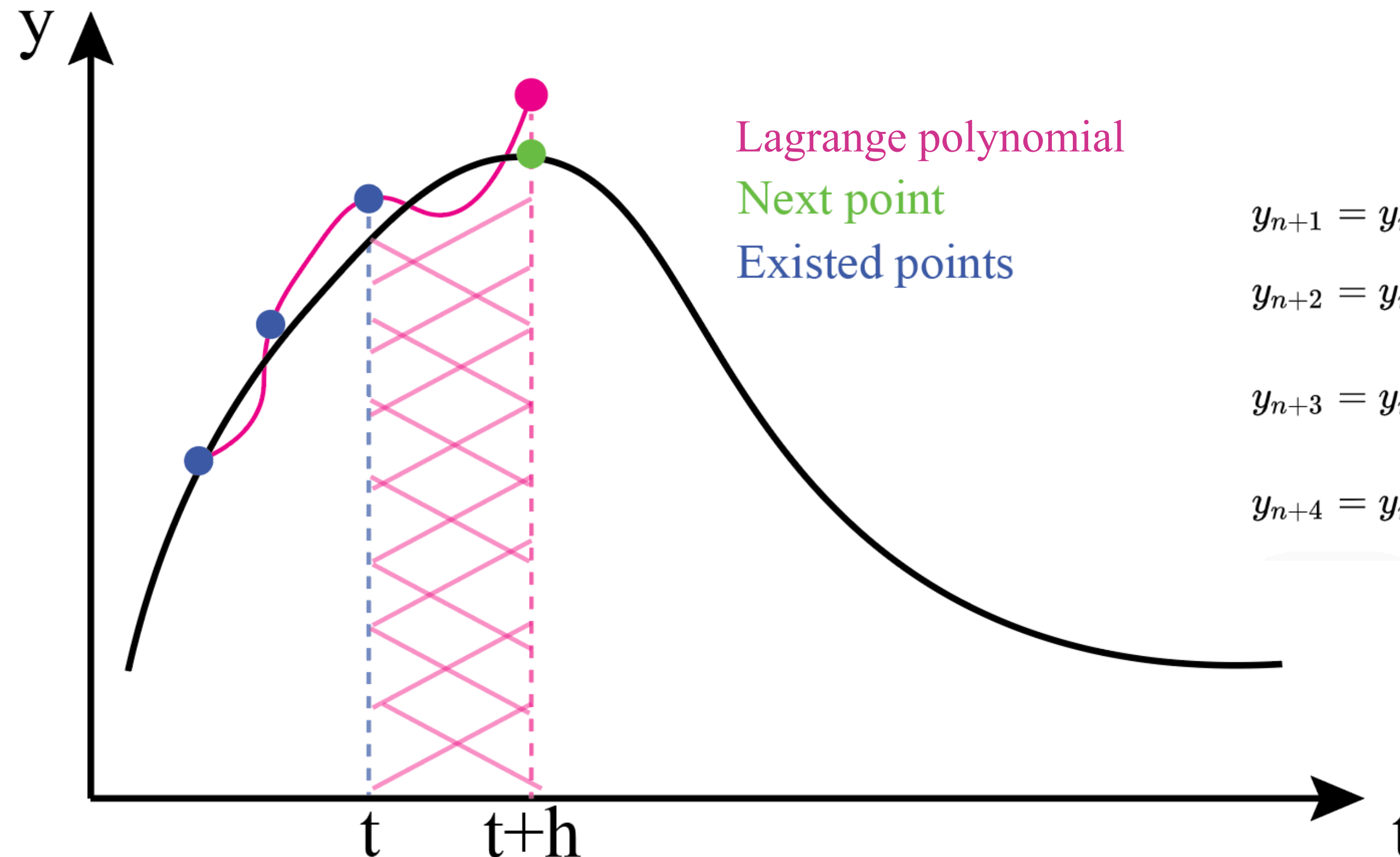
Adams-Bashforth methods (Multistep explicit methods)

Use past solution values and past derivative evaluations to compute the next step

$$y_{i+1} = y_i + \int_{t_i}^{t_{i+1}} f(t, y(t)) dt \quad \text{Cannot integrate } f(t, y(t))$$

$$f(t, y(t)) \approx p_{s-1}(t), \quad p_{s-1}(t) = \sum_{j=0}^{s-1} \frac{(-1)^{s-j-1} f(t_{n+j}, y_{n+j})}{j!(s-j-1)! h^{s-1}} \prod_{\substack{i=0 \\ i \neq j}}^{s-1} (t - t_{n+i})$$

Lagrange polynomial



$$y_{n+1} = y_n + h f(t_n, y_n), \quad (\text{This is the Euler method})$$

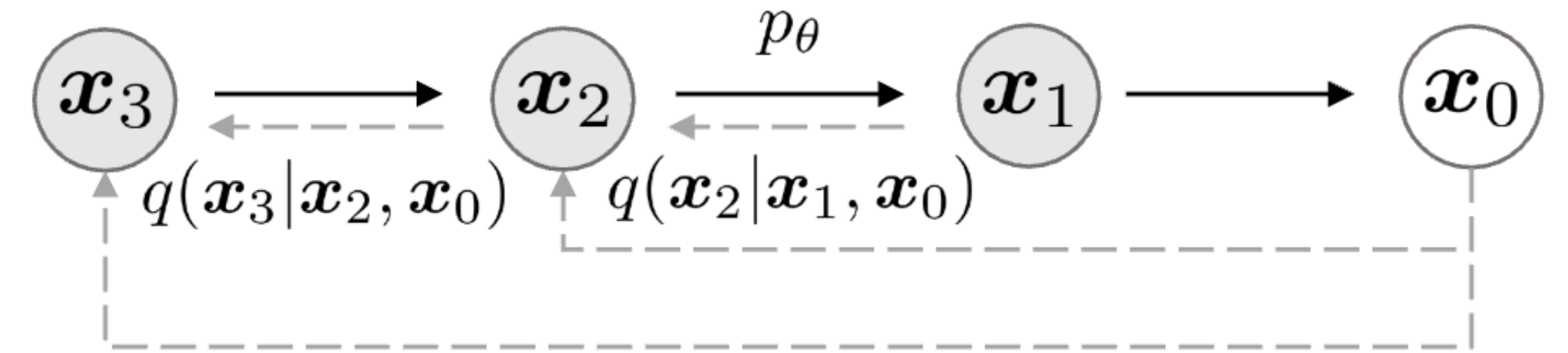
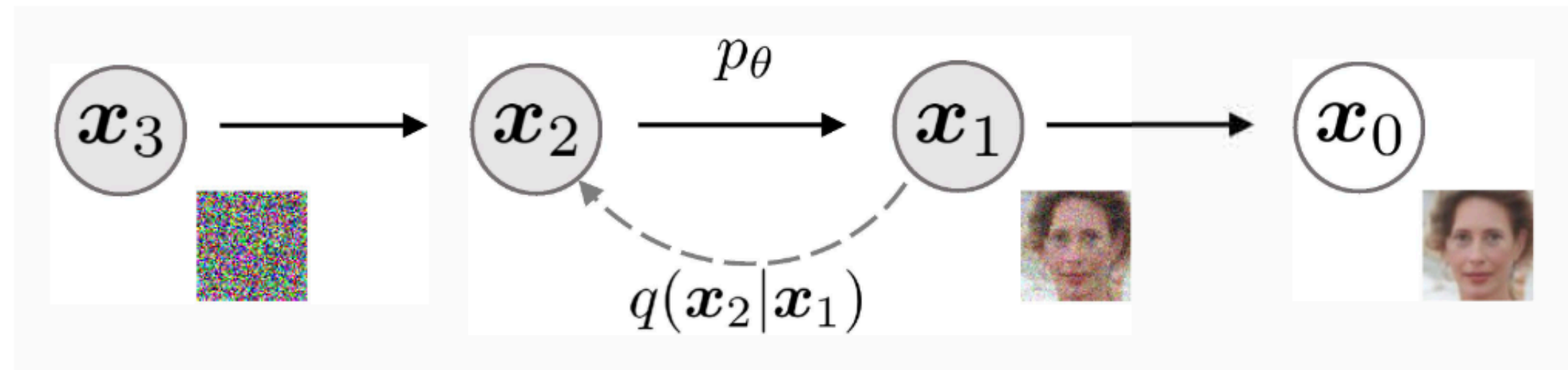
$$y_{n+2} = y_{n+1} + h \left(\frac{3}{2} f(t_{n+1}, y_{n+1}) - \frac{1}{2} f(t_n, y_n) \right),$$

$$y_{n+3} = y_{n+2} + h \left(\frac{23}{12} f(t_{n+2}, y_{n+2}) - \frac{16}{12} f(t_{n+1}, y_{n+1}) + \frac{5}{12} f(t_n, y_n) \right),$$

$$y_{n+4} = y_{n+3} + h \left(\frac{55}{24} f(t_{n+3}, y_{n+3}) - \frac{59}{24} f(t_{n+2}, y_{n+2}) + \frac{37}{24} f(t_{n+1}, y_{n+1}) - \frac{9}{24} f(t_n, y_n) \right),$$

3. DPM-solver (VP case)

DDIM (Denoising Diffusion Implicit Models)



$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon} \quad \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \approx \boldsymbol{\epsilon} \quad \hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}{\sqrt{\alpha_t}}$$

$$\mathbf{x}_t = \sqrt{\alpha_t} \left(\frac{\mathbf{x}_s - \sqrt{1 - \alpha_s} \boldsymbol{\epsilon}_\theta(\mathbf{x}_s, s)}{\sqrt{\alpha_s}} \right) + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_s, s)$$

3. DPM-solver (VP case)

DDIM (Denoising Diffusion Implicit Models)

$$\mathbf{x}_t = \sqrt{\alpha_t} \left(\frac{\mathbf{x}_s - \sqrt{1 - \alpha_s} \boldsymbol{\epsilon}_\theta(\mathbf{x}_s, s)}{\sqrt{\alpha_s}} \right) + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_s, s)$$

$$d\mathbf{x} = \frac{1}{2\alpha_t} \left[\mathbf{x} + \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] d\alpha_t$$

What is DDIM? Just Euler?

3. DPM-solver (VP case)

DDIM (Denoising Diffusion Implicit Models)

$$\mathbf{x}_t = \sqrt{\alpha_t} \left(\frac{\mathbf{x}_s - \sqrt{1 - \alpha_s} \boldsymbol{\epsilon}_\theta(\mathbf{x}_s, s)}{\sqrt{\alpha_s}} \right) + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_s, s)$$

$$d\mathbf{x} = \frac{1}{2\alpha_t} [\mathbf{x} + \nabla_{\mathbf{x}} \log p_t(\mathbf{x})] d\alpha_t$$

What is DDIM? Just Euler?

$$d\mathbf{x} = \frac{1}{2\alpha_t} \left[\mathbf{x} - \frac{\boldsymbol{\epsilon}(\mathbf{x}, t)}{\sqrt{1 - \alpha_t}} \right] d\alpha_t$$

$$\mathbf{x}_t = \frac{\alpha_t + \alpha_s}{2\alpha_s} \mathbf{x}_s - \frac{\alpha_t - \alpha_s}{2\alpha_s \sqrt{1 - \alpha_s}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_s, s)$$

$$\mathbf{x}_t = \mathbf{x}_s + \int_s^t \frac{1}{2\alpha_\tau} \left[\mathbf{x} - \frac{\boldsymbol{\epsilon}(\mathbf{x}, \tau)}{\sqrt{1 - \alpha_t}} \right] d\alpha_\tau$$

$$d\mathbf{x} = \frac{1}{2\alpha_\tau} \left[\mathbf{x} - \frac{\epsilon(\mathbf{x}, \tau)}{\sqrt{1 - \alpha_\tau}} \right] d\alpha_\tau$$

$$d\mathbf{x} = \left[\frac{\mathbf{x}}{2\alpha_\tau} - \frac{\epsilon(\mathbf{x}, \tau)}{2\alpha_\tau \sqrt{1 - \alpha_\tau}} \right] d\alpha_t, s \rightarrow t$$

$$\phi(t, \tau) d\mathbf{x} = \phi(t, \tau) \left[\frac{\mathbf{x}}{2\alpha_\tau} - \frac{\epsilon(\mathbf{x}, \tau)}{2\alpha_\tau \sqrt{1 - \alpha_\tau}} \right] d\alpha_\tau,$$

$$d \left[\mathbf{x} \phi(t, \tau) \right] = - \phi(t, \tau) \frac{\epsilon(\mathbf{x}, \tau)}{2\alpha_\tau \sqrt{1 - \alpha_\tau}} d\alpha_\tau,$$

$$d\phi(t, \tau) = - \phi(t, \tau) \frac{1}{2\alpha_t} d\alpha_\tau$$

$$\int_s^t \frac{d\phi(t, \tau)}{\phi(t, \tau)} = - \int_s^t \frac{d\alpha_\tau}{2\alpha_\tau} \rightarrow \ln \phi(t, s) = \frac{1}{2} (\ln \alpha_t - \ln \alpha_s)$$

$$\phi(t, s) = \sqrt{\frac{\alpha_t}{\alpha_s}}$$

$$\mathbf{x}_t = \sqrt{\frac{\alpha_t}{\alpha_s}} \mathbf{x}_s - \frac{1}{2} \int_s^t \phi(t, \tau) \frac{\epsilon(\mathbf{x}, \tau)}{\alpha_\tau \sqrt{1 - \alpha_\tau}} d\alpha_\tau,$$

$$\mathbf{x}_t = \sqrt{\frac{\alpha_t}{\alpha_s}} \mathbf{x}_s - \frac{\sqrt{\alpha_t}}{2} \int_s^t \frac{\epsilon(\mathbf{x}, \tau)}{\sqrt{\alpha_\tau} \alpha_\tau \sqrt{1 - \alpha_\tau}} d\alpha_\tau,$$

$$\int_s^t \frac{\epsilon(\mathbf{x}, \tau)}{\sqrt{\alpha_\tau} \alpha_\tau \sqrt{1 - \alpha_\tau}} d\alpha_\tau \approx \epsilon(\mathbf{x}_s, s) \int_s^t \frac{1}{\sqrt{\alpha_\tau} \alpha_\tau \sqrt{1 - \alpha_\tau}} d\alpha_\tau$$

$$\int \frac{d\alpha}{\sqrt{\alpha}\alpha\sqrt{1-\alpha}}$$

$$1-\alpha=\cos^2u,\quad \alpha=\sin^2u,\quad d\alpha=2\sin u\cos udu$$

$$\int \frac{2du}{\sin^2u}=-2\operatorname{ctg}u=-2\operatorname{ctg}\arcsin\sqrt{\alpha}=-2\frac{\sqrt{1-\alpha}}{\sqrt{\alpha}}$$

DDIM = DPM-1

$$\boldsymbol{x}_t=\sqrt{\frac{\alpha_t}{\alpha_s}}\boldsymbol{x}_s-\frac{\sqrt{\alpha_t}}{2}\left(-2\frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}+2\frac{\sqrt{1-\alpha_s}}{\sqrt{\alpha_s}}\right)\boldsymbol{\epsilon}(\boldsymbol{x}_s,s)$$

$$\boldsymbol{x}_t=\sqrt{\alpha_t}\left(\frac{\boldsymbol{x}_s-\sqrt{1-\alpha_s}\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_s,s)}{\sqrt{\alpha_s}}\right)+\sqrt{1-\alpha_t}\boldsymbol{\epsilon}(\boldsymbol{x}_s,s)$$

$$\mathbf{x}_t = \sqrt{\frac{\alpha_t}{\alpha_s}} \mathbf{x}_s - \frac{\sqrt{\alpha_t}}{2} \int_s^t \frac{\epsilon(\mathbf{x}, \tau)}{\sqrt{\alpha_\tau} \alpha_\tau \sqrt{1 - \alpha_\tau}} d\alpha_\tau,$$

$$\epsilon(\mathbf{x}, \tau) = \sum_{k=0}^{n-1} \frac{(\alpha_\tau - \alpha_s)^k}{k!} \epsilon^{(k)}(\mathbf{x}_s, s)$$

Singlestep (use the intermediate point)

$$\epsilon^{(1)}(\mathbf{x}_s, s) \approx \frac{\epsilon(\mathbf{x}_{s-\delta_s}, s - \delta_s) - \epsilon(\mathbf{x}_s, s)}{\alpha_{s-\delta_s} - \alpha_s}$$

Multistep (use the previous point)

$$\epsilon^{(1)}(\mathbf{x}_s, s) \approx \frac{\epsilon(\mathbf{x}_s, s) - \epsilon(\mathbf{x}_{s+h}, s + h)}{\alpha_s - \alpha_{s+h}}$$

DPM = Exponential integrator + Taylor Series methods

Exponential integrator + Adams–Bashforth ???

4. SDE solvers

$$d\mathbf{x} = \left[f(t)\mathbf{x} - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt + g(t) d\mathbf{w}_t$$

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \int_{t_i}^{t_{i+1}} \left(f(t)\mathbf{x}(t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t)) \right) dt + \int_{t_i}^{t_{i+1}} g(t) d\mathbf{W}_t$$

$$\int_{t_i}^{t_{i+1}} \left(f(t)\mathbf{x}(t) - g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}(t)) \right) dt = h \left(f(t_i)\mathbf{x}_i - g(t_i)^2 \nabla_{\mathbf{x}} \log p_{t_i}(\mathbf{x}_i) \right) + O(h^2)$$

$$\int_{t_i}^{t_{i+1}} g(t) d\mathbf{W}_t = g(t_i) \Delta \mathbf{W}_i + O(|h|^{3/2}), \quad \Delta \mathbf{W}_i = \sqrt{|h|} \mathbf{z}_i \sim \mathcal{N}(0, |h|I), \quad \mathbf{z}_i \sim \mathcal{N}(0, I)$$

Euler-Maruyama

$$\mathbf{x}_{i+1} = \mathbf{x}_i + h \left(f(t_i)\mathbf{x}_i - g(t_i)^2 \nabla_{\mathbf{x}} \log p_{t_i}(\mathbf{x}_i) \right) + g(t_i) \sqrt{|h|} \mathbf{z}_i$$

Reverse sampling: $h = t_{i+1} - t_i < 0$. Keep h signed in the drift; noise covariance is $|h|I$.

4. SDE solvers

Lower convergence

Harder higher-order

Noise increases variance of error

Need many steps

4. Appendix

$$\begin{aligned} dm_t &= f(t)m_tdt, \; m_0 = x_0 \\ dv_t &= \left(2f(t)v_t + g^2(t)\right) dt, \; v_0 = 0 \end{aligned}$$

Path 1. SDE — > Marginals

$$f(t) = \frac{1}{2\alpha_t} \frac{d\alpha_t}{dt}, \; g^2(t) = -\frac{1}{\alpha_t} \frac{d\alpha_t}{dt}$$

$$dm_t = \frac{1}{2\alpha_t} \frac{d\alpha_t}{dt} m_t dt, \; m_0 = x_0$$

$$\ln \frac{m_t}{m_0} = \frac{1}{2} \ln \frac{\alpha_t}{\alpha_0}$$

$$dm_t = \frac{1}{2\alpha_t} d\alpha_t m_t$$

$$m_t = x_0 \sqrt{\alpha_t}$$

$$\int_0^t \frac{dm_t}{m_t} = \int_0^t \frac{1}{2} \frac{d\alpha_t}{\alpha_t}$$

4. Appendix

$$dm_t = f(t)m_tdt, m_0 = x_0$$
$$dv_t = \left(2f(t)v_t + g^2(t)\right) dt, v_0 = 0$$

Path 1. SDE —> Marginals

$$f(t) = \frac{1}{2\alpha_t} \frac{d\alpha_t}{dt}, g^2(t) = -\frac{1}{\alpha_t} \frac{d\alpha_t}{dt}$$

$$dv_t = \left(2\frac{1}{2\alpha_t} \frac{d\alpha_t}{dt} v_t - \frac{1}{\alpha_t} \frac{d\alpha_t}{dt}\right) dt, v_0 = 0$$

$$dv_t = \left(\frac{1}{\alpha_t} v_t - \frac{1}{\alpha_t}\right) d\alpha_t$$

$$\frac{dv_t}{v_t - 1} = \frac{d\alpha_t}{\alpha_t}$$

$$\ln \frac{v_t - 1}{v_0 - 1} = \ln \frac{\alpha_t}{\alpha_0}$$

$$v_t - 1 = (v_0 - 1)\alpha_t$$

$$v_t = 1 - \alpha_t$$

4. Appendix

$$\begin{aligned} dm_t &= f(t)m_t dt, \quad m_0 = x_0 \\ dv_t &= \left(2f(t)v_t + g^2(t) \right) dt, \quad v_0 = 0 \end{aligned}$$

Path 2. Marginal —> SDE

$$x_t = x_0 \sqrt{\alpha_t} + \sqrt{1 - \alpha_t} z$$

$$\begin{aligned} dx_0 \sqrt{\alpha_t} &= f(t)x_0 \sqrt{\alpha_t} dt, & d(1 - \alpha_t) &= \left(2 \frac{1}{2\alpha_t} \frac{d\alpha_t}{dt} (1 - \alpha_t) + g^2(t) \right) dt \\ f(t) &= \frac{1}{2\alpha_t} \frac{d\alpha_t}{dt} & g(t)^2 &= -\frac{1}{\alpha_t} \frac{d\alpha_t}{dt} \end{aligned}$$